CBSE MOCK — Pre-Board Practice Series

Central Board of Secondary Education

CLASS X — SCIENCE

Date: 12 November 2026 | Maximum Marks: 80 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 5 sections. Read the instructions for each section carefully.
2. All questions are compulsory unless stated otherwise.
3. Write all answers neatly and show relevant working wherever required.
4. Marks for each question are indicated against it.
5. This paper is a Practice Paper — it is similar to but NOT an actual CBSE board paper.
6. Download more FREE practice papers daily at: **www.cbsemock.org.in**

SECTION A — Biology (30 Marks)

A/R (1 mark each) | LA (5 marks each) | SA (3 marks each) | VSA (2 marks each) | MCQ (1 mark each) | CBQ (4 marks each — Case Study)

Q1. **[3 marks]**

Study the given table and answer the following questions: Parents: A pea plant with red flowers was crossed with a pea plant with white flowers. F_1: 30 seeds sown, all bore red flowers. F_2: 44 seeds sown, 33 plants bore red flowers and 11 plants bore white flowers. (a) Depict the information given in the table in the form of a cross. (b) Deduce the F_2 phenotypic ratio/F_2 ratio showing external look as well as genotypic/genetic makeup ratio.

Q2. **[1 mark]**

Assertion (A): Highest percentage of pesticides is found in tertiary consumers in a food chain. Reason (R): Chemicals/pesticides get accumulated progressively at each trophic level.

Q3. **[1 mark]**

Identify the traits that cannot be transmitted from parents to the offsprings: (i) Swimming technique (ii) Blood group (iii) Eye colour (iv) Dancing skills

(A) (i) and (ii)
(B) (i) and (iv)
(C) (iii) and (iv)
(D) (ii) and (iii)

Q4. **[1 mark]**

Given below are some statements regarding our environment. Which amongst these is not leading to any damage of environment? (i) Change in life-styles have resulted in greater amounts of waste material generation. (ii) Changes in attitude have resulted in use of more disposable things. (iii) Changes in packaging has resulted in much of our waste becoming non-biodegradable. (iv) Organic farming.

(A) (i), (ii), (iii)
(B) (iv)
(C) (i) and (iii)
(D) (ii) and (iv)

Q5. **[3 marks]**

Give reasons for the following: (a) Adrenal gland is known as the stress gland of the body. (b) Prenatal sex determination is banned in India. (c) Vegetative propagation is advantageous in some plants.

Q6. **[2 marks]**

Explain why food chains generally consist of three or four steps.

Q7. **[2 marks]**

(A) Varun is diabetic. His latest ultrasound reports showed a blockage in his pancreatic duct. How will this affect the dual functioning of the pancreas? OR (B) Sunil unconsciously touched the hot iron rod and immediately withdrew his hand. Which type of action is involved? Also define this action. Draw a flow chart to show the path followed for this action.

Questions 8 & 9 consist of two statements: Assertion (A) and Reason (R). Select appropriate option from (A),(B),(C),(D): (A) Both A and R are true and R is the correct explanation of A. (B) Both A and R are true, but R is not the correct explanation of A. (C) A is true, but R is false. (D) A is false, but R is true.

Q8. **[1 mark]**

Assertion (A): Variations are not often seen in offsprings produced by asexual reproduction. Reason (R): DNA molecule generated by replication is not similar to the original DNA.

Q9. **[2 marks]**

"Transpiration in plants is a necessary evil." Comment on the statement giving two reasons.

Q10. **[4 marks]**

Plants exhibit movements in response to various environmental stimuli to adopt and survive in their surroundings. These movements, influenced by external factors such as light, gravity, water and chemicals are regulated by specific plant hormones. For example, the differential growth of plant parts allows them to bend, grow or orient themselves in particular directions. Roots and shoots often show contrasting response to the same stimulus, highlighting the complex mechanisms involved in plant growth and development. (a) Why is hydrotropism essential for plants? (b) What role does auxin play in the growth of plants? (c) Differentiate between movement shown by the sensitive plant and tropic movement by plants (Two points). OR (c) How does movement of roots differ from that of shoots in response to gravity and light?

Q11. **[1 mark]**

Following are the statements about the different parts of the brain. Select the correct statements. (i) The thinking part of the brain is the forebrain. (ii) Centres for smell, hearing, memory, sight are located in forebrain. (iii) Salivation, vomiting, blood pressure are controlled by medulla in the hind brain. (iv) Cerebellum does not control the posture and balance of the body.

(A) (i), (ii) and (iii)
(B) (ii), (iii) and (iv)
(C) (i), (ii) and (iv)
(D) (i), (iii) and (iv)

Q12. **[1 mark]**

In human beings the oxygen rich blood is pumped out to the body by:

(A) contraction of right atrium
(B) relaxation of left ventricle
(C) contraction of right ventricle
(D) contraction of left ventricle

Q13. **[1 mark]**

If the part 'P' of the flower is removed, which of the following processes will not take place?

(A) Formation of pollens
(B) Formation of flower
(C) Formation of fruit
(D) Formation of seed

Q14. **[1 mark]**

Two pea plants, one with round green seeds (RRyy) and another with wrinkled yellow (rrYY) seeds were crossed with each other which produced F_1 progeny with round yellow (RrYy) seeds. When F_1 plants are self-pollinated, the F_2 progeny will have which new combination of characters, as compared to the parents?

(i) Round yellow (ii) Round green (iii) Wrinkled yellow (iv) Wrinkled green

(A) (i) and (ii)
(B) (i) and (iv)
(C) (ii) and (iv)
(D) (i) and (iii)

Q15. **[5 marks]**

(A) (a) Draw a labelled diagram showing open and closed stomatal pore. (b) List the main events that occur during the process of photosynthesis. (c) Write a balanced chemical equation showing photosynthesis in plants. OR (B) (a) Draw a well-labelled diagram of nephron and label at least 4 of its parts. (b) Write the role of capillary cluster inside Bowman's capsule. (c) What is meant by selective reabsorption?

Q16. **[1 mark]**

Observe the given figure and select the statement that correctly states how these fungi obtain nutrition.

(A) By eating the bread on which it is growing.
(B) By using nutrients from the bread to prepare their own food.
(C) By breaking down the nutrients of bread outside the body and then absorbing them.
(D) By allowing other organisms to grow on the bread and then consuming them.

SECTION B — Chemistry (25 Marks)

A/R (1 mark each) | SA (3 marks each) | VSA (2 marks each) | MCQ (1 mark each) | CBQ (4 marks each — Case Study)

Q17. **[2 marks]**

A shiny metal X on exposure to air turns dull. Name the phenomenon and the compound responsible for this change. Also name metal X.

Q18. **[3 marks]**

A student added a few drops of universal indicator to three solutions A, B and C. The colour changed to orange for A, blue for B and red for C. (a) Arrange A, B and C in increasing order of their pH values. (b) Which solution is strongly acidic? (c) If sodium hydroxide is added to solution A, what change would you observe?

Q19. **[2 marks]**

Write the balanced chemical equation for the reaction of: (a) Iron with steam (b) Zinc with dilute sulphuric acid

Q20. **[1 mark]**

An aqueous solution turns red litmus blue. Excess addition of which of the following solution would reverse the change?

(A) Sodium hydroxide
(B) Hydrochloric acid
(C) Baking soda
(D) Slaked lime

Q21. **[4 marks]**

Carbon and its compounds form the basis of all life processes. Carbon forms a vast number of compounds due to its unique property of catenation. It also shows tetravalency. The carbon compounds are of two types — hydrocarbons and their derivatives. (a) What is catenation? Why does silicon show less tendency of catenation than carbon? (b) Write the IUPAC name of CH_3-CH_2-CH(OH)-CH_3. (c) Write the structural formula of Ethanoic acid. OR (c) Write the name and formula of the functional group present in alcohols. Also name the product formed when ethanol is heated with excess of conc. H_2SO_4 at 170°C.

Q22. **[3 marks]**

List the observations made when: (a) HCl is added to Na_2CO_3 solution (b) NaOH solution is added to $CuSO_4$ solution (c) Zinc granules are added to dilute H_2SO_4

Q23. **[1 mark]**

The chemical formula of washing soda is:

(A) Na_2CO_3
(B) $NaHCO_3$
(C) $Na_2CO_3 \cdot 10H_2O$
(D) NaOH

Q24. **[1 mark]**

Assertion (A): Respiration is an exothermic process. Reason (R): Glucose is oxidised to release energy during respiration.

Q25. **[1 mark]**

Assertion (A): Copper does not react with dilute sulphuric acid. Reason (R): Copper is less reactive than hydrogen in the activity series.

Q26. **[1 mark]**

While washing dirty clothes by soap, micelles are formed. Select the correct statement about micelles.

(A) The hydrophobic tail of the soap molecule is in the interior of the cluster, whereas hydrophilic end is on the surface.
(B) The hydrophobic tail is on the surface of the cluster, whereas hydrophilic end is in the interior.
(C) Both hydrophobic tail and hydrophilic end are in the interior of the cluster.
(D) Both hydrophobic tail and hydrophilic end are on the surface of the cluster.

Q27. **[1 mark]**

The volume ratio of hydrogen and oxygen gases liberated during electrolysis of water is:

(A) 1:1
(B) 2:1
(C) 4:1
(D) 1:2

Q28. **[1 mark]**

Which of the following metals reacts with cold water to form a metal hydroxide and hydrogen gas?

(A) Magnesium
(B) Zinc
(C) Sodium
(D) Iron

Q29. **[1 mark]**

$C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$. The above reaction is an example of:

(A) Displacement Reaction
(B) Endothermic Reaction
(C) Exothermic Reaction
(D) Neutralisation Reaction

SECTION C — Physics (25 Marks)

A/R (1 mark each) | SA (3 marks each) | VSA (2 marks each) | MCQ (1 mark each) | CBQ (4 marks each — Case Study)

Q30. **[2 marks]**

(A) Why is the energy of the sun considered to be inexhaustible? Name the nuclear reactions responsible for the energy of the sun. OR (B) With the help of a labelled diagram, explain how a step-up transformer works.

Q31. **[3 marks]**

A student performs an experiment with a convex lens. He places a candle at different positions and observes the image on a screen. Complete the following table: (a) Object at infinity — Image position and nature? (b) Object between F and 2F — Image position and nature? (c) Object at F — Image position and nature?

Q32. **[1 mark]**

If the current through a conductor is doubled, the drift velocity of the electrons will:

(A) remain the same
(B) become half
(C) double
(D) become four times

Q33. **[1 mark]**

A concave mirror has a focal length of 15 cm. An object is placed 45 cm in front of the mirror. The image is formed at:

(A) 22.5 cm in front
(B) 22.5 cm behind
(C) 45 cm behind
(D) 45 cm in front

Q34. **[1 mark]**

Which of the following phenomena is responsible for the blue colour of the sky?

(A) Reflection of light
(B) Refraction of light
(C) Scattering of light
(D) Total internal reflection

Q35. **[2 marks]**

Draw the magnetic field lines due to a bar magnet and state the two properties of magnetic field lines.

Q36. **[1 mark]**

The SI unit of electric power is:

(A) Volt
(B) Ampere
(C) Ohm
(D) Watt

Q37. **[1 mark]**

Assertion (A): A convex lens always forms a real image of a real object. Reason (R): A convex lens has the ability to converge a beam of light.

Q38. **[1 mark]**

A ray of light passes from a denser medium to a rarer medium. At the interface, the angle of refraction is 90°. The angle of incidence is called:

(A) Angle of reflection
(B) Critical angle
(C) Angle of dispersion
(D) Angle of deviation

Q39. **[1 mark]**

In a series circuit, three resistors R_1, R_2 and R_3 are connected. If $R_1 = 2\Omega$, $R_2 = 3\Omega$ and $R_3 = 5\Omega$, the total resistance is:

(A) 1 Ω
(B) 10 Ω
(C) 15 Ω
(D) 30 Ω

Q40. **[4 marks]**

Ohm's law states that the potential difference V across the ends of a given metallic wire at constant temperature is directly proportional to the current I flowing through it. The constant of proportionality is called resistance R. The resistance of a wire depends on its length, cross-sectional area, and the material (resistivity). (a) State the formula for resistivity and write its SI unit. (b) A wire of resistance 4Ω is stretched to double its length. What will be the new resistance? (c) Why are alloys commonly used in electrical heating devices rather than pure metals? OR (c) Three resistors of 2Ω, 4Ω and 6Ω are connected in parallel. Find the equivalent resistance.

Q41. **[2 marks]**

An electric bulb is rated 220V–100W. When it is connected to a 110V supply, what will be the power consumed?

Q42. **[1 mark]**

Assertion (A): Electromagnets are made of soft iron. Reason (R): Soft iron has high retentivity and low coercivity.

✦ END OF QUESTION PAPER ✦

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